Project Proposal ON

Prodify: AI-Powered Focus & Burnout Tracking

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Introduction

In today's fast-paced digital environment, maintaining focus and managing burnout are critical challenges for students and professionals. Traditional productivity tools often fail to adapt to individual work habits or lack predictive insights regarding fatigue. The integration of Artificial Intelligence (AI) and Machine Learning (ML) provides an opportunity to address these challenges. By leveraging AI, individuals can forecast burnout risk, identify distraction patterns, and optimize their work-rest cycles.

This project focuses on developing a comprehensive system called **Prodify** that combines real-time intent tracking with burnout prediction. Prodify is designed to help users track their focus and predict burnout in real-time. Its Windows Tauri desktop client monitors the active application and window titles, compares that activity with a user's workspace plan, and feeds the resulting focus data into a live fatigue-risk model. An integrated AI Coaching assistant offers personalized, data-driven advice, making it a valuable tool for sustainable productivity.

Objectives

- **Design an intuitive and user-friendly desktop application** capable of tracking productivity sessions, helping users stay aligned with their goals.
- **Leverage advanced machine learning techniques** to uncover productivity patterns and predict burnout probabilities based on live and historical telemetry data.
- **Offer tailored, real-time AI coaching** to enhance user well-being and maximize focus periods, while encouraging personalized break strategies.
- **Streamline distraction tracking** by implementing consecutive detection timers and OS-level application monitoring to minimize non-productive context switching.
- **Develop interactive dashboards** to present actionable insights in an engaging and visually appealing manner, facilitating data-driven decision-making for long-term task management.

Scope

This project is designed to address the challenges faced by digital workers in managing focus and preventing burnout. The system combines AI-powered coaching with ML-based burnout prediction to provide actionable insights and optimized strategies.

- **Pomodoro & Focus Tracking:** A robust finite state machine (FSM) implementation for

structured focus and break sessions.
- **Desktop Intent & Distraction Tracking:** Real-time monitoring of tab-switching and foreground applications to identify distractions and enforce focus via visual interventions.
- **Live Burnout Prediction:** Utilize predictive models on historical telemetry data to calculate burnout risk dynamically during active sessions.
- **AI Coaching Assistant:** Incorporate real-time conversational AI to offer personalized productivity advice based on the user's live session data.
- **Broad Applicability:** The system can be scaled from student study routines to professional software development workflows, making it flexible and versatile.

Methodology

The development of the Prodify system will follow a structured and phased approach, ensuring the delivery of a robust and functional solution. The methodology includes the following key stages:

- **Requirements Analysis:** Identify the key objectives, such as burnout forecasting, focus tracking, and AI integration. Understand the data requirements and scope of the system to meet productivity needs.
- **Data Collection and Preprocessing:** Log telemetry data (focus duration, distraction counts, session states) during user sessions. Preprocess the data by extracting session features and standardizing metrics for analysis.
- **Machine Learning Implementation:** Apply predictive algorithms (such as RandomForestClassifier) to forecast burnout probability based on engineered features. Evaluate the model's accuracy.
- **System Design and Development:**
    - **Frontend:** Design an intuitive and user-friendly interface using React and Tailwind CSS, bundled with Tauri for desktop capabilities.
    - **Backend:** Use FastAPI (Python) to handle data processing, run ML models, and integrate with the Groq API for AI capabilities.
    - **Database:** Implement SQLite for local data persistence of User, Workspace, Session, and Telemetry profiles.
- **Testing and Evaluation:** Conduct unit and integration testing to ensure all components work seamlessly together. Test system performance with real-world sessions to validate predictions and usability.
- **Deployment:** Package the Tauri desktop application for Windows accessibility and run the backend server processes. Ensure proper documentation for users.

Tools and Technologies

- **Languages:** Python (ML and Backend), TypeScript/HTML/CSS (Frontend), Rust (Tauri).
- **Libraries/Frameworks:**
    - **Machine Learning:** Scikit-learn, Pandas, Joblib.
    - **Frontend:** React, Tailwind CSS, Zustand (State Management).
    - **Backend:** FastAPI, Uvicorn, SQLAlchemy.
    - **Desktop Client:** Tauri.
    - **AI Platform:** Groq API (Llama-3.3-70b-versatile).

- **Database:** SQLite.

Resources

1. Software Resources

- **Python:** For backend logic and Machine Learning (Scikit-learn, FastAPI).
- **Node.js / React:** For handling user interfaces and state.
- **Rust / Tauri Toolchain:** For cross-platform desktop application packaging.
- **SQLite:** For storing transactional session and telemetry data.

2. System Requirements

- **Hardware:**
  - Processor: Intel i5 or higher (or AMD equivalent).
  - RAM: 8GB or more.
  - Storage: 500MB free space.
- **Software:**
  - Windows 10/11 OS.
  - Python 3.8+ and Node.js 18+.

3. Dataset Resources

- **Telemetry Logs:** Self-generated synthetic and real-time usage data (focus duration, time since last break, distraction events per session).

Expected Outcome

- Users can actively monitor their productivity patterns and focus duration through continuous usage.
- Graphical representations of burnout risk using predictive analysis will provide enhanced well-being awareness.
- Facilitates an enforced workflow that blocks distractions before they significantly impact a session.
- Provides dynamic AI coaching interventions, offering timely suggestions to enhance scholastic and professional output while avoiding fatigue.

References

[1] Documentation and research on AI-enhanced productivity tracking. [2] "Predicting Burnout in Software Engineering" - Research context for machine learning applied to workflow metrics. [3] Tauri framework and FastAPI documentation for building hybrid desktop-web applications.